
Eco-Link
A SOLAR-POWERED PISO WIFI SYSTEM
with Integrated Coin Capacity Alert
STD
SOFTWARE TEST DOCUMENT
ASTRYX
ECO-LINK
ECO-LINK
124
₱1,245.00
75%
Coin Box Capacity
ECO-LINK
75%
Recent Alerts
High Coin Capacity Alert
₱245.00

ECO-LINK

A Solar-Powered Piso WiFi System with Integrated Coin Capacity Alert

SOFTWARE TEST DOCUMENT (STD)

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1. Introduction

1.1 Purpose

This Software Test Document (STD) defines the complete testing strategy, test plans, test cases, and test procedures for the Eco-Link system — a Solar-Powered Piso WiFi vending machine with an Integrated Coin Capacity Alert. It provides the authoritative test reference for all Quality Assurance activities, ensuring that every functional and non-functional requirement derived from the project's Review of Related Literature (RRL) is systematically verified and validated before deployment.

This STD is structured in accordance with IEEE Std 829-2008 and ISO/IEC/IEEE 29119-3 and is intended to guide testing across all project phases: unit testing, integration testing, system testing, and user acceptance testing (UAT).

1.2 Project Background

The Eco-Link system addresses documented limitations in existing Piso WiFi infrastructure in the Philippines. Based on the RRL, the key gaps targeted are:

- Lack of sustainable, off-grid power sourcing for Piso WiFi units in remote barangays (Safaei et al., 2025; Kumar et al., 2024)
- Absence of proactive coin-holder capacity alerts, causing machine downtime and lost operator revenue (Bose et al., 2025; Nimbalkar et al., 2025)
- Weak network security in existing public WiFi deployments (Akbar et al., 2025; Shafana & Ibrahim, 2026)
- No centralized remote monitoring for multi-unit operators (Petilla, 2025; Parmar et al., 2024)

1.3 Scope of Testing

Testing will cover the following system modules and integration points:

Module	Description	Test Levels Applied
Solar Power Management	PV harvesting, battery SOC, source switching, fault isolation	Unit, Integration, System
Coin Capacity Monitoring	Sensor reading, fill-level computation, threshold alerts	Unit, Integration, System
Captive Portal & Sessions	WiFi redirect, coin-triggered activation, session expiry	Integration, System, UAT
Network Security	RADIUS, TLS, RBAC, certificate validation, brute-force protection	System, Security, Penetration
Cloud Dashboard	Real-time telemetry display, remote configuration, historical charts	System, Performance, UAT
Push Notification Service	Alert delivery, retry logic, operator preference management	Integration, System
OTA Firmware Update	Remote firmware delivery and rollback	System

Module	Description	Test Levels Applied
Data Consistency & Pipeline	MQTT pipeline, InfluxDB storage, sync-on-reconnect	Integration, System, Performance

1.4 Definitions, Acronyms, and Abbreviations

Term	Definition
STD	Software Test Document
SRS	Software Requirements Specification
RRL	Review of Related Literature
UAT	User Acceptance Testing
IEEE 829	IEEE Standard for Software and System Test Documentation
SOC	State of Charge (battery level)
RADIUS	Remote Authentication Dial-In User Service
LDAP	Lightweight Directory Access Protocol
TLS	Transport Layer Security
MitM	Man-in-the-Middle attack
RBAC	Role-Based Access Control
FIDO2	Fast IDentity Online 2 – passwordless authentication standard
MQTT	Message Queuing Telemetry Transport (IoT protocol)
InfluxDB	Open-source time-series database for telemetry data
OTA	Over-the-Air (firmware update)
ESP32	Dual-core Wi-Fi enabled microcontroller used as IoT node
FCM	Firebase Cloud Messaging (push notification service)
DUT	Device Under Test
TC	Test Case
Pass/Fail	Test case outcome: Pass = expected result met; Fail = deviation observed

1.5 References

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2. Test Plan Overview

2.1 Test Objectives

The primary objectives of this test effort are to:

- Verify that all functional requirements defined in the SRS are correctly implemented and behave as specified
- Validate that the system performs reliably under expected and stressed operating conditions
- Confirm that all security mechanisms are resistant to known attack vectors (brute-force, MitM, certificate spoofing)
- Ensure that the solar power management module correctly handles SOC transitions, source switching, and fault isolation without data loss
- Confirm that push notifications are delivered within acceptable latency under normal and degraded network conditions
- Validate that the system meets scalability targets for up to 50 concurrent Eco-Link units on a single dashboard

2.2 Test Strategy

Test Level	Approach	Tools / Methods
Unit Testing	Test individual firmware functions and software modules in isolation	PlatformIO unit test framework, Jest (Node.js), Python unittest
Integration Testing	Verify data flow across hardware-software boundaries (sensor → ESP32 → MQTT → InfluxDB)	Postman, MQTT Explorer, Wireshark
System Testing	End-to-end testing of all modules operating together on the physical hardware	Manual test execution, automated scripts
Performance Testing	Measure telemetry throughput, dashboard latency, CPU overhead under load	Locust (load testing), Grafana dashboards
Security Testing	Penetration testing of captive portal, RADIUS, and TLS configurations	Hashcat, Nmap, OpenSSL CLI, Metasploit (controlled)
User Acceptance Testing (UAT)	Operator and end-user validation in a real-world deployment scenario	Manual walkthrough with stakeholders

2.3 Test Entry and Exit Criteria

2.3.1 Entry Criteria

- All hardware components are assembled and functional (solar panel, battery, ESP32, sensors, router)
- Firmware v1.0 is flashed on the ESP32 DUT
- Cloud infrastructure (MQTT broker, InfluxDB, dashboard, FCM) is deployed and accessible

- Test environment network is isolated from production
- All test cases in this STD have been reviewed and approved by the project lead

2.3.2 Exit Criteria

- All High-priority test cases have been executed and achieve Pass status
- Zero open Critical or Blocker defects remain unresolved
- All Medium-priority test cases have been executed; up to 5% may be deferred with documented justification
- Security testing shows no exploitable vulnerabilities in RADIUS, TLS, or captive portal
- UAT sign-off obtained from at least one machine operator stakeholder

2.4 Test Environment

Component	Specification	Purpose
ESP32 DevKit V1	Dual-core 240 MHz, 4MB Flash, built-in Wi-Fi	Primary IoT DUT (Device Under Test)
Solar Panel (Test Rig)	12V 30W monocrystalline PV panel	Simulate real-world energy input
LiFePO4 Battery Bank	12V 20Ah, BMS-protected	Energy storage DUT
INA219 Current Sensor	I2C, 26V/3.2A, calibrated to ±1%	Power monitoring accuracy testing
HC-SR04 Ultrasonic	2–50cm range, ±3mm accuracy	Coin level sensor DUT
MikroTik hAP ac²	Dual-band router, RouterOS v7.x, RADIUS client	Captive portal and network DUT
InfluxDB 2.x Server	Cloud-hosted instance, Flux query	Telemetry storage backend
Grafana 10.x	Connected to InfluxDB datasource	Dashboard visualization DUT
Wireshark 4.x	Packet capture on test VLAN	Network traffic analysis
OpenSSL CLI	v3.x	TLS certificate validation testing
Postman v11	API test collections	REST API endpoint testing
Android Test Device	Android 13, Chrome 120+	Captive portal mobile UX testing

2.5 Risk and Mitigation

Risk	Likelihood	Impact	Mitigation
Inconsistent solar input during outdoor tests	High	Medium	Use a DC bench power supply set to 12–18V to simulate PV output deterministically
ESP32 firmware crashes during stress tests	Medium	High	Implement watchdog timer; log crash dumps via serial monitor
False-positive coin sensor readings	Medium	Medium	Calibrate sensor in controlled environment before each test session
Push notification delivery delays (FCM)	Low	Medium	Mock FCM server for latency-sensitive test cases; test real FCM separately
RADIUS server unavailable during security tests	Low	High	Deploy local FreeRADIUS instance in test environment as fallback

3. Test Cases — Module 1: Solar Power Management

Literature Basis: Safaei et al. (2025), Kumar et al. (2024), Petilla (2025), Prakash (2026), Rahman et al. (2025)

3.1 Unit Tests — SOC Monitoring and Source Switching

TC ID	Test Case Name	Precondition	Test Steps	Expected Result	Priority	Status
TC-SPM-01	Solar voltage continuous monitoring	ESP32 powered; INA219 connected to solar input	1. Supply 14.5V to solar input via bench supply.2. Read INA219 via I2C.3. Verify firmware reads voltage every ≤10 seconds.	Voltage reading logged at ≤10s intervals; value within ±0.1V of bench supply output.	High	Pass
TC-SPM-02	Hysteresis SOC switching: solar to battery	Battery at 80% SOC; solar input active at 13V	1. Reduce solar input below 11.5V threshold.2. Monitor power source status flag in firmware.	System switches to battery source within 3 seconds; no relay hunting observed.	High	Pass
TC-SPM-03	Hysteresis SOC switching: battery to solar	Battery at 60% SOC; solar input OFF	1. Apply solar input voltage of 13.5V.2. Monitor source switching behavior.	System re-activates solar source; hysteresis prevents switch-back for ≥30 seconds unless solar drops again.	High	Pass
TC-SPM-04	Deep discharge prevention at 20% SOC	Solar OFF; battery draining under load	1. Simulate discharge until SOC = 21%.2. Continue draining to 20%.3. Monitor relay state.	Load relay opens at exactly 20% SOC; system logs disconnect event with timestamp.	High	Pass
TC-SPM-05	Auto-reconnect at 30% SOC recovery	Load disconnected at 20% SOC	1. Charge battery to 30% SOC via solar.2. Monitor relay state.	Load relay closes automatically at 30% SOC; system resumes normal operation.	High	Pass
TC-SPM-06	SOC threshold: edge case at exactly 20%	Battery draining	1. Precisely simulate SOC = 20.0%.2. Observe relay behavior.	Load relay opens; no oscillation (hysteresis enforced). Single disconnect event logged.	Medium	Pass

3.2 Integration Tests — Power Telemetry to Cloud

TC ID	Test Case Name	Precondition	Test Steps	Expected Result	Priority	Status
TC-EPT-01	Telemetry publish to MQTT broker	ESP32 connected to Wi-Fi; MQTT broker online	1. Power on ESP32.2. Monitor MQTT topic 'ecolink/{unit_id}/power'.3. Wait 65 seconds.	At least one telemetry packet published within 60 seconds. Payload includes: solar_v, battery_soc, active_source, timestamp.	**High**	**Pass**
TC-EPT-02	Fault detection: overvoltage event	Normal operation active	1. Apply 15.5V to battery input (simulated overvoltage).2. Monitor relay and alert.	Relay opens within 2 seconds. Overvoltage event published to MQTT. Dashboard shows fault status.	**High**	**Pass**
TC-EPT-03	Fault detection: undervoltage event	Normal operation active	1. Drop solar input to 9V.2. Monitor fault flag.	Undervoltage fault detected within 2 seconds. Load isolated. Event logged to InfluxDB with fault code.	**High**	**Pass**
TC-EPT-04	Low battery operator alert at 30% SOC	Battery discharging; dashboard logged in	1. Discharge to 31% SOC.2. Continue to 30%.3. Monitor notification.	Push notification delivered to operator device within 30 seconds. Alert includes: unit ID, SOC value, timestamp.	**High**	**Pass**
TC-EPT-05	30-day telemetry data retention in InfluxDB	InfluxDB bucket configured with 30d retention	1. Inject 31-day-old test records.2. Query records at day 30 and day 31.	Day 30 records returned; day 31 records absent (auto-expired by retention policy).	**Medium**	**N/A**
TC-EPT-06	Telemetry recovery after internet outage	Normal telemetry active	1. Disconnect internet for 5 minutes.2. Reconnect.3. Check InfluxDB.	All telemetry packets buffered locally during outage. All packets synced to InfluxDB within 60 seconds of reconnect.	**High**	**Pass**

4. Test Cases — Module 2: Coin Capacity Monitoring

Literature Basis: Bose et al. (2025), Nimbalkar et al. (2025), Flores et al. (2025)

4.1 Unit Tests — Coin Level Sensor

TC ID	Test Case Name	Precondition	Test Steps	Expected Result	Priority	Status
TC-CCM-01	Sensor calibration: empty coin holder	HC-SR04 mounted; coin holder empty	1. Trigger sensor reading.2. Record raw distance value.3. Set as 0% calibration baseline.	Distance reading stable (±3mm). Firmware stores as D_MAX (empty distance).	**High**	**Pass**
TC-CCM-02	Sensor calibration: full coin holder	Coin holder loaded to physical maximum	1. Trigger sensor reading.2. Record raw distance value.3. Set as 100% calibration baseline.	Distance reading stable (±3mm). Firmware stores as D_MIN (full distance).	**High**	**Pass**
TC-CCM-03	Fill percentage computation accuracy	Calibration completed (TC-CCM-01, 02)	1. Load coin holder to 50% by count.2. Trigger fill percentage reading.	Firmware reports 48–52% fill (within ±2% accuracy band).	**High**	**Pass**
TC-CCM-04	Sensor polling frequency	ESP32 firmware running	1. Enable serial debug log.2. Observe coin sensor read events over 5 minutes.	Sensor polled at least once every 10 seconds (minimum 30 readings in 5 minutes).	**High**	**Pass**
TC-CCM-05	Anomalous reading detection	Sensor obstructed mid-air (out-of-range)	1. Place hand in front of sensor at 1cm (below minimum range).2. Observe firmware response.	Firmware flags reading as anomalous; does not update fill percentage; logs anomaly event.	**Medium**	**Pass**

4.2 Integration Tests — Alert Thresholds and Notifications

TC ID	Test Case Name	Precondition	Test Steps	Expected Result	Priority	Status
TC-CCM-06	Warning push notification at 85% fill	Operator app registered; FCM configured	1. Fill coin holder to 84%.2. Add coins until sensor reads 85%.3. Monitor notifications.	Push notification delivered to operator within 30 seconds. Message: 'Unit [ID] coin holder at 85% capacity. Please collect soon.'	**High**	**Pass**
TC-CCM-07	Critical push notification at 95% fill	Operator app registered; machine accepting coins	1. Fill to 94%.2. Add coins to reach 95%.3. Monitor coin acceptance and alerts.	Critical push notification delivered within 30 seconds. Machine stops accepting new coins. Dashboard shows FULL status.	**High**	**Pass**
TC-CCM-08	Machine resumes after operator reset	Coin holder at 95%; machine stopped	1. Operator empties coin holder.2. Press physical reset button.3. Monitor machine state.	Machine resumes accepting coins. Coin counter resets to 0%. Dashboard updates to 0% within 60 seconds.	**High**	**Pass**
TC-CCM-09	Remote reset via dashboard	Dashboard logged in as Operator role; coin holder full	1. Navigate to unit management on dashboard.2. Click 'Reset Coin Counter'.3. Monitor machine state.	Coin counter resets remotely. Machine accepts coins. Dashboard reflects new state within 5 seconds.	**High**	**Pass**
TC-CCM-10	Notification not re-triggered on stable reading	Coin holder at exactly 85%	1. Keep coin holder at 85% (no additional coins) for 5 minutes.2. Monitor notifications.	No duplicate 85% warning notifications are sent. Notification fires once per threshold crossing, not repeatedly.	**Medium**	**Pass**
TC-CCM-11	Coin insertion event log accuracy	System operational; coin holder < 50% full	1. Insert 20 coins manually.2. Check insertion event log.	Log contains 20 insertion events, each with timestamp and cumulative count. No events missed.	**Medium**	**Pass**

5. Test Cases — Module 3: Captive Portal & Network Security

Literature Basis: Akbar et al. (2025), Rivera-Dourado et al. (2026), Shafana & Ibrahim (2026)

5.1 Captive Portal Functional Tests

TC ID	Test Case Name	Precondition	Test Steps	Expected Result	Priority	Status
TC-CPM-01	Captive portal redirect on WiFi connect	MikroTik router configured; hotspot enabled	1. Connect test Android device to Eco-Link SSID.2. Open browser.3. Attempt to navigate to any URL.	Browser automatically redirects to captive portal page. No internet access before portal interaction.	High	Pass
TC-CPM-02	Session activated by coin insertion	User on captive portal page; no active session	1. Insert 1 peso coin into hardware.2. Monitor user's browser.	Session activated within 5 seconds of coin detection. User gains internet access. Timer starts.	High	Pass
TC-CPM-03	Session expiry and re-redirect	Active session with 15-minute duration purchased	1. Allow session to run to expiry.2. Monitor user's browser after expiry.	Internet access revoked. User redirected to captive portal. Timer shows 00:00.	High	Pass
TC-CPM-04	Session tier display on portal page	Captive portal accessible	1. Open captive portal on test device.2. Inspect displayed tier information.	Portal displays available tiers (e.g., 1 peso = 15 min, 5 pesos = 1 hr) clearly and correctly.	High	Pass
TC-CPM-05	Maximum concurrent users enforcement	10-user limit configured on router	1. Connect 10 devices and activate sessions.2. Connect 11th device.	First 10 devices have active sessions. 11th device sees captive portal with 'Maximum users reached' message.	Medium	Pass
TC-CPM-06	Captive portal loads within 3 seconds on mobile	Android test device on WiFi; portal not cached	1. Clear browser cache.2. Connect to Eco-Link SSID.3. Measure portal load time.	Captive portal fully rendered within 3 seconds from first redirect.	Medium	Pass

TC ID	Test Case Name	Precondition	Test Steps	Expected Result	Priority	Status
TC-CPM-07	Portal accessible without app installation	Android 13 device, Chrome 120 browser	1. Connect fresh device (no apps installed).2. Connect to SSID.3. Interact with portal.	Captive portal functions fully in mobile browser. No app download required.	**High**	**Pass**

5.2 Network Security Tests

TC ID	Test Case Name	Precondition	Test Steps	Expected Result	Priority	Status
TC-SEC-01	RADIUS authentication: valid credentials	FreeRADIUS configured; test user created	1. Attempt management VLAN login with valid credentials.2. Monitor RADIUS log.	Access granted. RADIUS log shows Accept message with timestamp and username.	**High**	**Pass**
TC-SEC-02	RADIUS authentication: invalid credentials	FreeRADIUS configured	1. Attempt login with wrong password 3 times.2. Monitor RADIUS log.	Access denied. RADIUS log shows Reject message. No management interface access granted.	**High**	**Pass**
TC-SEC-03	Brute-force lockout after 5 failures	Dashboard login page accessible	1. Attempt login with wrong password 5 consecutive times.2. Attempt 6th login.3. Wait 15 minutes.	Account locked after 5th failure. Alert sent to admin. Login resumes after 15-minute lockout.	**High**	**Pass**
TC-SEC-04	TLS 1.2+ enforcement on all HTTPS endpoints	OpenSSL CLI available	1. Run: openssl s_client -connect dashboard:4432. Inspect negotiated protocol version.	TLS 1.2 or 1.3 negotiated. Connection successful. No TLS 1.0 or 1.1 accepted.	**High**	**Pass**
TC-SEC-05	SSL/TLS certificate chain validation at startup	Misconfigured test certificate loaded (incomplete chain)	1. Start system with incomplete cert chain.2. Monitor startup logs and dashboard.	Startup diagnostic detects cert chain error. Critical alert generated. System proceeds but flags cert warning.	**High**	**Pass**

TC ID	Test Case Name	Precondition	Test Steps	Expected Result	Priority	Status
TC-SEC-06	MitM detection via certificate mismatch	Wireshark on test VLAN	1. Inject self-signed cert at network level.2. Monitor captive portal TLS handshake.	Cert mismatch detected by validation engine. User sees warning indicator. Event logged with cert chain details.	**High**	**Pass**
TC-SEC-07	RBAC: Operator cannot access Admin functions	Operator-role user logged into dashboard	1. Log in as Operator role.2. Attempt to access Admin-only settings (e.g., RADIUS config).	Admin section returns 403 Forbidden. Operator cannot view or modify admin settings.	**Medium**	**Pass**
TC-SEC-08	RBAC: Read-Only cannot modify configurations	Read-Only user logged into dashboard	1. Log in as Read-Only.2. Attempt to change any dashboard setting.	All configuration controls are disabled/hidden. No changes possible. No error thrown on attempt.	**Medium**	**Pass**
TC-SEC-09	Dashboard session timeout at 30 minutes idle	Operator logged into dashboard	1. Log in and leave dashboard idle for 30 minutes.2. Attempt any action.	Session expired. User redirected to login page. No dashboard data accessible without re-authentication.	**Medium**	**Pass**
TC-SEC-10	Password hashing validation (bcrypt/Argon2)	Access to authentication database	1. Create test user with known password.2. Inspect stored password hash in database.	Stored value is a bcrypt or Argon2 hash (not plaintext). Hash prefix verifiable (e.g., $2b$ for bcrypt).	**High**	**Pass**

6. Test Cases — Module 4: Cloud Dashboard & Notifications

Literature Basis: Petilla (2025), Parmar et al. (2024), Sambandam (2023), Kitao & Ikeda (2016)

6.1 Dashboard Functional Tests

TC ID	Test Case Name	Precondition	Test Steps	Expected Result	Priority	Status
TC-DSH-01	Real-time unit overview display	2 Eco-Link units registered and online	1. Log into dashboard as Operator.2. Open unit overview page.3. Observe data for both units.	Both units displayed with: status (Online/Offline), battery SOC %, coin fill %, active session count. Data refreshes automatically.	**High**	**Pass**
TC-DSH-02	Dashboard telemetry latency ≤5 seconds	Unit actively publishing telemetry	1. Physically change battery SOC by connecting load.2. Record timestamp of change.3. Record timestamp of dashboard update.	Dashboard reflects new SOC value within 5 seconds of physical change.	**Medium**	**Pass**
TC-DSH-03	Historical trend chart: 24-hour view	24+ hours of telemetry data in InfluxDB	1. Select unit on dashboard.2. Open trend chart.3. Select '24h' time range.	Chart renders power and coin level trends for past 24 hours. Data points present for every 10-minute interval.	**Medium**	**Pass**
TC-DSH-04	Historical trend chart: 7-day view	7+ days of telemetry data	1. Select '7d' time range on chart.	Chart renders 7-day trend. Axes correct. No missing data gaps > 60 minutes.	**Medium**	**Pass**
TC-DSH-05	Remote alert threshold configuration	Operator logged in; default thresholds active	1. Navigate to unit settings.2. Change coin warning threshold from 85% to 75%.3. Save.4. Fill coin holder to 76%.	Alert fires at 75% fill on next check cycle. New threshold persists after ESP32 restart.	**High**	**Pass**
TC-DSH-06	Multi-unit dashboard: 10 units simultaneously	10 Eco-Link units configured (7 real + 3 simulated)	1. Register 10 units to dashboard.2. Monitor dashboard with all units publishing simultaneously.	All 10 units display correctly. No units missing or stuck. Dashboard	**Medium**	**Pass**

TC ID	Test Case Name	Precondition	Test Steps	Expected Result	Priority	Status
				responsive (<3s page load).		

6.2 Push Notification Service Tests

TC ID	Test Case Name	Precondition	Test Steps	Expected Result	Priority	Status
TC-NTF-01	Notification content accuracy	Operator phone registered; FCM configured	1. Trigger a coin-85% alert.2. Inspect notification payload on phone.	Notification contains: unit ID, event type ('Coin 85% Full'), current sensor reading, UTC timestamp.	High	Pass
TC-NTF-02	Notification delivery for all 4 event types	All event types configured; operator phone registered	1. Trigger each event: coin 85%, coin 95% (critical), battery low, unit offline.2. Monitor phone.	4 distinct notifications received within 60 seconds each. All contain correct event-specific content.	High	Pass
TC-NTF-03	Retry logic on failed delivery	FCM endpoint temporarily mocked to fail	1. Block FCM endpoint for 1 minute.2. Trigger coin-85% alert.3. Re-enable FCM endpoint.	System retries delivery up to 3 times with exponential backoff. Final delivery succeeds on re-enable.	Medium	Pass
TC-NTF-04	Operator disables specific notification type	Operator logged into dashboard	1. Navigate to notification preferences.2. Disable 'Battery Low' notification.3. Trigger battery low condition.	No push notification sent for battery low event. All other notification types still delivered normally.	Medium	Pass
TC-NTF-05	No duplicate notifications for single event	Coin holder at stable 85%	1. Hold coin holder at exactly 85% for 10 minutes.2. Count notifications received.	Only 1 notification received for the single threshold crossing. No repeated alerts.	Medium	Pass

7. Performance Tests

Literature Basis: Sambandam (2023), Parmar et al. (2024), Rivera-Dourado et al. (2026), Rahman et al. (2025)

TC ID	Test Case Name	Method	Target Metric	Pass Threshold	Priority	Status
TC-PERF-01	Telemetry throughput under sustained operation	Run system for 24 hours; count delivered vs. expected MQTT packets	Packet delivery rate	≥99% of expected packets delivered (864/864 in 24h @ 60s interval)	High	**Pass**
TC-PERF-02	Fault response time	Trigger overvoltage; measure time from detection to relay open	Relay response latency	Relay opens within 2 seconds of fault detection in 10/10 trials	High	**Pass**
TC-PERF-03	Dashboard latency under 10-unit load	10 units publishing simultaneously; measure dashboard page response	Page response time	Dashboard loads within 3 seconds under 10-unit telemetry load	Medium	**Pass**
TC-PERF-04	ESP32 CPU overhead from telemetry	Profile firmware CPU usage with and without telemetry active	Telemetry CPU overhead	Telemetry collection consumes ≤5% additional CPU (per Sambandam, 2023 cross-node model)	Medium	**Pass**
TC-PERF-05	Captive portal load time on mobile	Load portal 10 times on test Android; measure from redirect to fully rendered	Portal render time	Average render time ≤3 seconds across 10 trials; no trial exceeds 5 seconds	Medium	**Pass**
TC-PERF-06	Battery backup duration	Disconnect solar; monitor load runtime until SOC hits 20% cutoff	Backup runtime	Load operates for minimum 8 hours on battery alone from 100% SOC to cutoff	High	**Pass**
TC-PERF-07	System uptime over 30-day simulated operation	Operate system in lab for 30 days; track uptime events	Monthly uptime percentage	≥95% uptime (maximum 36 hours downtime per 30-day period)	High	**N/A**

8. Security Penetration Tests

Literature Basis: Akbar et al. (2025) — RADIUS + LDAP penetration testing with Hashcat; Shafana & Ibrahim (2026) — SSL/TLS diagnostic; Rivera-Dourado et al. (2026) — FIDO2/WebAuthn limitations

Note: All penetration tests are conducted in an isolated test environment on hardware owned by the project team. No production systems or third-party networks are involved.

TC ID	Test Case Name	Attack Vector	Test Method	Pass Condition	Priority	Status
TC-PEN-01	Brute-force resistance on dashboard login	Credential brute-force	Use Hashcat wordlist attack against login endpoint after capturing challenge hash. Attempt 10,000 passwords.	No valid credential discovered. Account lockout engages after 5 attempts. High-entropy hash not cracked.	High	**Pass**
TC-PEN-02	MitM certificate spoofing detection	SSL/TLS interception	Deploy rogue AP with self-signed cert; attempt to intercept captive portal handshake using SSLstrip.	Validation engine detects certificate chain mismatch. Alert generated. User warned. Session not established.	High	**Pass**
TC-PEN-03	RADIUS bypass attempt via MAC spoofing	MAC address forgery	Clone MAC of authorized device; attempt management VLAN access without valid RADIUS credentials.	RADIUS rejects access: MAC filtering alone insufficient without valid credentials. Access denied.	High	**Pass**
TC-PEN-04	Captive portal session hijacking attempt	Cookie/token theft	Capture session cookie via Wireshark; replay token from unauthorized device.	Session bound to MAC address. Replayed token rejected. Unauthorized device not granted access.	High	**Pass**
TC-PEN-05	MQTT unauthorized publish attempt	Unauthorized IoT message injection	Attempt to publish to 'ecolink/#' MQTT topic without TLS client certificate.	MQTT broker rejects connection. Unauthenticated client cannot publish or	High	**Pass**

TC ID	Test Case Name	Attack Vector	Test Method	Pass Condition	Priority	Status
				subscribe to system topics.		
TC-PEN-06	Dashboard SQL/NoSQL injection attempt	Input injection	Submit injection payloads (' OR 1=1--, {$gt: ''}) in all dashboard input fields.	No database errors returned. All inputs sanitized. No unauthorized data disclosure or authentication bypass.	High	**Pass**
TC-PEN-07	TLS downgrade attack resistance	Protocol downgrade	Attempt TLS connection forcing TLS 1.0 via OpenSSL: openssl s_client -tls1 -connect dashboard:443	Server rejects TLS 1.0 handshake. Connection refused. Only TLS 1.2+ accepted.	High	**Pass**

9. User Acceptance Testing (UAT)

UAT is conducted with real stakeholders in a near-production environment. Participants include at least one machine operator and two end-user volunteers.

9.1 UAT Scenarios — Machine Operator

UAT ID	Scenario	User Action	Acceptance Criteria	Participant	Result
UAT-OP-01	First-time unit registration	Operator registers a new Eco-Link unit using the setup guide	Registration completed within 15 minutes; unit appears on dashboard with live telemetry	Machine Operator A	**Pass**
UAT-OP-02	Receive and respond to coin-full alert	Coin holder filled to 95%; operator receives push notification and empties machine	Notification received within 60 seconds; remote reset via dashboard works; machine accepts coins again	Machine Operator A	**Pass**
UAT-OP-03	View power telemetry on dashboard	Operator opens dashboard and reviews battery SOC and solar voltage history for past 7 days	Charts load within 5 seconds; data present and clearly labeled; operator understands reading without training	Machine Operator A	**Pass**
UAT-OP-04	Configure custom alert thresholds	Operator changes coin warning threshold from 85% to 70% via dashboard	Threshold saved; system triggers alert at 70% on next test; no technical support needed by operator	Machine Operator A	**Pass**
UAT-OP-05	Manage multiple units on single account	Operator views 3 registered units simultaneously on dashboard overview	All 3 units displayed with correct individual readings; operator can identify which unit needs attention	Machine Operator A	**Pass**

9.2 UAT Scenarios — End User (WiFi Customer)

UAT ID	Scenario	User Action	Acceptance Criteria	Participant	Result
UAT-EU-01	Connect and purchase WiFi session	User connects to Eco-Link SSID, sees captive portal, inserts 1 peso, and browses the internet	Internet access granted within 10 seconds of coin insertion; no instructions needed beyond portal text	End User Volunteer 1	**Pass**
UAT-EU-02	Session expiry awareness	User’s session expires; they re-purchase via portal	User redirected to portal at expiry; re-purchase works without confusion; no technical errors shown	End User Volunteer 1	**Pass**
UAT-EU-03	Portal usability on mobile (Filipino language)	User reads and interacts with portal displayed in Filipino	User understands all portal text and completes session purchase without assistance	End User Volunteer 2	**Pass**
UAT-EU-04	Connection with older Android device (Android 9)	User connects older device to Eco-Link SSID	Captive portal loads correctly on Android 9; session activates normally; no app required	End User Volunteer 2	**Pass**

10. Data Consistency and Pipeline Tests

Literature Basis: Kitao & Ikeda (2016) — Sequential/Causal consistency, PACELC theorem; Parmar et al. (2024) — InfluxDB time-series pipeline; Sambandam (2023) — cross-node telemetry overhead reduction

TC ID	Test Case Name	Test Steps	Expected Result	Priority	Status
TC-DATA-01	MQTT message ordering integrity	1. Publish 100 telemetry packets in sequence with incrementing sequence numbers.2. Verify order in InfluxDB.	All 100 packets stored in InfluxDB in correct timestamp order. No out-of-order records.	High	Pass
TC-DATA-02	No data loss during network partition (5-min outage)	1. Disconnect ESP32 internet for 5 minutes while telemetry active.2. Reconnect.3. Count InfluxDB records.	Local buffer retains all packets during outage. All packets synced post-reconnect. Zero records lost.	High	Pass
TC-DATA-03	Concurrent multi-unit data isolation	1. Publish telemetry from Unit A and Unit B simultaneously with similar timestamps.2. Query InfluxDB by unit_id.	Data correctly segregated by unit_id tag. No cross-unit data contamination.	High	Pass
TC-DATA-04	InfluxDB quorum read consistency	1. Write a telemetry record.2. Immediately read it back via Flux query.	Record returned on immediate read. No stale-read observed (eventual consistency lag < 1 second).	Medium	Pass
TC-DATA-05	Telemetry CPU overhead per Sambandam model	1. Profile ESP32 CPU with telemetry disabled.2. Enable telemetry.3. Measure delta CPU usage over 10 minutes.	Telemetry overhead ≤5% of total CPU. No telemetry-induced latency spikes on coin or power processing.	Medium	Pass
TC-DATA-06	30-day data retention policy enforcement	1. Insert records with timestamp = today − 31 days.2. Query for those records after InfluxDB runs retention policy.	Records older than 30 days purged automatically. Records at exactly 30 days retained.	Medium	N/A

11. Ethical and Compliance Tests

Literature Basis: RRL Ethical Considerations section; Data Privacy Act of 2012 (R.A. 10173); Flores et al. (2025) on environmental responsibility; Nimbalkar et al. (2025) on inclusive design

TC ID	Test Case Name	Test Steps	Expected Result	Priority	Status
TC-ETH-01	Minimum data collection verification	1. Inspect all data stored in InfluxDB and session logs.2. Check for any PII beyond MAC address and session duration.	No names, phone numbers, emails, or device identifiers beyond MAC address stored anywhere in the system.	High	Pass
TC-ETH-02	Data encryption at rest	1. Access InfluxDB storage volume directly.2. Attempt to read raw telemetry files.	Data on disk is encrypted (AES-256). Raw files are not human-readable without decryption key.	High	Pass
TC-ETH-03	Session log auto-purge at 90 days	1. Insert session log records dated 91 days ago.2. Wait for purge cycle.3. Query for old records.	Records older than 90 days purged. Records at exactly 90 days retained.	Medium	N/A
TC-ETH-04	Coin-based access: inclusive payment validation	1. Test session activation using only coin insertion (no digital payment).2. Complete a full session.	Session successfully activated by coin alone. No requirement for digital payment, app, or phone number.	High	Pass
TC-ETH-05	Portal text in Filipino language	1. Open captive portal.2. Verify Filipino translation is present and grammatically correct.	All user-facing portal text available in Filipino. Translation verified by native Filipino speaker.	Medium	Pass
TC-ETH-06	Solar energy primary power verification	1. Disconnect grid power (if any).2. Operate system under solar and battery only for 8 hours.	System operates fully for 8 hours on solar+battery. No grid connection required. Carbon-free operation confirmed.	High	Pass
TC-ETH-07	No user content filtering or deep packet inspection	1. Route test device traffic through Eco-Link.2. Inspect traffic logs for content inspection signatures.	No DPI signatures or content filtering rules active beyond session enforcement. User browsing is private.	Medium	Pass

12. OTA Firmware Update Tests

Literature Basis: Bose et al. (2025) — self-diagnostic and proactive maintenance model for IoT vending machines

TC ID	Test Case Name	Test Steps	Expected Result	Priority	Status
TC-OTA-01	Successful OTA firmware update	1. Upload v1.1 firmware to OTA server.2. Trigger update from dashboard.3. Monitor ESP32 reboot.	ESP32 downloads, verifies (SHA256 checksum), and installs v1.1 firmware. Reboots and reports new version within 3 minutes.	**High**	**Pass**
TC-OTA-02	OTA update during active WiFi sessions	1. Establish 3 active user sessions.2. Trigger OTA update.	Update queued until sessions expire OR users warned of upcoming maintenance. Sessions not abruptly terminated without notice.	**Medium**	**Pass**
TC-OTA-03	Rollback on corrupted firmware package	1. Upload intentionally corrupted firmware (invalid checksum).2. Trigger OTA update.	ESP32 detects checksum mismatch. Update aborted. Device rolls back to previous firmware version. Alert sent to admin.	**High**	**Pass**
TC-OTA-04	Self-diagnostic routine: 24-hour cycle	1. Operate system normally.2. Monitor dashboard at 24-hour intervals.	Dashboard receives hardware health report every 24 hours. Report includes: sensor status, relay status, connectivity status, firmware version.	**Medium**	**Pass**
TC-OTA-05	Self-diagnostic error code transmission	1. Disconnect HC-SR04 sensor.2. Trigger diagnostic cycle.3. Monitor dashboard.	Diagnostic detects sensor fault. Error code published to dashboard within 60 seconds. Admin alert generated with specific fault code identifying coin sensor.	**High**	**Pass**

13. Test Results Summary

The table below summarizes the overall test execution status across all modules and test levels. N/A indicates test cases requiring extended operation periods (30-day, 90-day) that are documented for post-deployment validation.

Module / Area	Total TCs	Pass	Fail	N/A	Pass Rate
Solar Power Management – Unit Tests	6	6	0	0	100%
Solar Power Management – Integration Tests	6	5	0	1	83% (1 N/A)
Coin Capacity Monitoring – Unit Tests	5	5	0	0	100%
Coin Capacity Monitoring – Alert Tests	6	6	0	0	100%
Captive Portal – Functional Tests	7	7	0	0	100%
Network Security Tests	10	10	0	0	100%
Cloud Dashboard – Functional Tests	6	6	0	0	100%
Push Notification Service Tests	5	5	0	0	100%
Performance Tests	7	6	0	1	86% (1 N/A)
Security Penetration Tests	7	7	0	0	100%
User Acceptance Testing (UAT)	9	9	0	0	100%
Data Consistency & Pipeline Tests	6	5	0	1	83% (1 N/A)
Ethical & Compliance Tests	7	6	0	1	86% (1 N/A)
OTA Firmware Update Tests	5	5	0	0	100%
— TOTAL —	92	88	0	4	96% active; 100% of executed TCs passed

Overall Assessment: All executed test cases PASSED. The 4 N/A cases are deferred to post-deployment observation phases (30-day uptime, 90-day log purge, 31-day retention). No critical or blocker defects were identified. The system is ready for UAT sign-off and staged production deployment.

14. Defect Log

The following table tracks all defects identified during test execution. This log is maintained throughout the testing lifecycle.

Defect ID	Related TC	Description	Severity	Status	Resolution
DEF-001	TC-CPM-06	Captive portal occasionally renders in 3.2–3.5s on Android 9 devices due to legacy TLS negotiation overhead	**Low**	**N/A**	Acceptable; within 5s hard limit. Optimize TLS cipher list in v1.1 firmware release.
DEF-002	TC-NTF-03	FCM retry delivers notification successfully but with 45-second delay on 3rd retry attempt	**Medium**	**N/A**	Delay is within acceptable retry policy. Retry backoff curve to be reviewed in next sprint.

No Critical or Blocker defects were found during system testing. Both logged defects are Low-Medium severity with documented workarounds and scheduled fixes in the next firmware version.

15. Requirements-to-Test Traceability Matrix

Each row maps a key requirement (from the SRS / RRL) to the test cases that verify it and the literature that supports the requirement.

Requirement / Feature	Literature Source	Test Case IDs
Continuous solar voltage monitoring	Kumar et al. (2024)	TC-SPM-01
Hysteresis SOC source switching	Kumar et al. (2024)	TC-SPM-02, TC-SPM-03, TC-SPM-06
Deep battery discharge prevention (20% cutoff)	Safaei et al. (2025)	TC-SPM-04
Auto-reconnect at 30% SOC	Safaei et al. (2025)	TC-SPM-05
Power telemetry to cloud (≤60s)	Petilla (2025)	TC-EPT-01
Fault detection & relay isolation (≤2s)	Prakash (2026)	TC-EPT-02, TC-EPT-03, TC-PERF-02
Low-battery alert at 30% SOC	Petilla (2025)	TC-EPT-04
Telemetry sync after internet outage	Parmar et al. (2024)	TC-EPT-06, TC-DATA-02
Coin sensor polling (≤10s)	Bose et al. (2025)	TC-CCM-01–05
85% coin warning push notification	Bose et al. (2025)	TC-CCM-06, TC-NTF-01, TC-NTF-02
95% coin critical alert + machine stop	Nimbalkar et al. (2025)	TC-CCM-07
Operator reset (physical + remote)	Nimbalkar et al. (2025)	TC-CCM-08, TC-CCM-09
Captive portal redirect on WiFi connect	Akbar et al. (2025)	TC-CPM-01
Coin-triggered session activation	Nimbalkar et al. (2025)	TC-CPM-02
Session expiry and re-redirect	Rivera-Dourado et al. (2026)	TC-CPM-03
Maximum concurrent user enforcement	Akbar et al. (2025)	TC-CPM-05
Portal load time ≤3 seconds	Rivera-Dourado et al. (2026)	TC-CPM-06, TC-PERF-05
Portal on mobile without app install	Rivera-Dourado et al. (2026)	TC-CPM-07, UAT-EU-01
RADIUS authentication and access control	Akbar et al. (2025)	TC-SEC-01, TC-SEC-02, TC-PEN-03
Brute-force lockout (5 attempts)	Akbar et al. (2025)	TC-SEC-03, TC-PEN-01

Requirement / Feature	Literature Source	Test Case IDs
TLS 1.2+ enforcement	Shafana & Ibrahim (2026)	TC-SEC-04, TC-PEN-07
SSL/TLS cert chain validation	Shafana & Ibrahim (2026)	TC-SEC-05, TC-PEN-02
RBAC: role-based dashboard access	Akbar et al. (2025)	TC-SEC-07, TC-SEC-08
Dashboard session timeout (30 min)	General Security Practice	TC-SEC-09
Password hashing (bcrypt/Argon2)	Akbar et al. (2025)	TC-SEC-10
Real-time multi-unit dashboard overview	Petilla (2025)	TC-DSH-01, TC-DSH-06, UAT-OP-01
Dashboard telemetry latency ≤5s	Petilla (2025)	TC-DSH-02
Historical trend charts (24h, 7d)	Parmar et al. (2024)	TC-DSH-03, TC-DSH-04
Remote threshold configuration	Petilla (2025)	TC-DSH-05, UAT-OP-04
Notification retry with exponential backoff	General Reliability Practice	TC-NTF-03
Telemetry CPU overhead ≤5%	Sambandam (2023)	TC-DATA-05, TC-PERF-04
Data ordering & consistency in InfluxDB	Kitao & Ikeda (2016)	TC-DATA-01, TC-DATA-03, TC-DATA-04
OTA firmware delivery and rollback	Bose et al. (2025)	TC-OTA-01, TC-OTA-03
Self-diagnostic with error code transmission	Bose et al. (2025)	TC-OTA-04, TC-OTA-05
Minimum data collection (Data Privacy Act)	R.A. 10173 (2012)	TC-ETH-01, TC-ETH-02
Coin-only inclusive access	Nimbalkar et al. (2025)	TC-ETH-04, UAT-EU-01
Solar as primary power (sustainability)	Flores et al. (2025)	TC-ETH-06, TC-PERF-06
95% monthly uptime target	Rahman et al. (2025)	TC-PERF-07 (post-deployment)

16. Test Completion and Sign-Off

The signatures below confirm that all applicable test cases have been executed, reviewed, and accepted, and that the Eco-Link system meets the agreed quality standards for deployment.

Role	Name	Signature	Date
QA Lead / Test Manager		________________________ ____	
Lead Developer		________________________ ____	
Hardware Engineer		________________________ ____	
Machine Operator (UAT)		________________________ ____	
Project Supervisor		________________________ ____	